CISS240: Introduction to Programming
Quiz q1703

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Q1. Write a C++ program that declares an array `a` of 1000 doubles *initializing* all the values to 0.0, obtain 1000 doubles from the user to be placed in the array (using a loop), compute and print the average of the values in the array (using another loop), and finally compute and print the average of the values in the array which are strictly positive (i.e., > 0) (using yet another loop).
Answer:

```
#include <iostream>

int main()
{
    double a[1000] = {0.0};

    for (int i = 0; i < 1000; ++i)
    {
        std::cin >> a[i];
    }

    double total_sum = 0.0;
    for (int i = 0; i < 1000; ++i)
    {
        total_sum += a[i];
    }
    double overall_average = total_sum / 1000.0;
    std::cout << overall_average << '\n';

    double positive_sum = 0.0;
    int positive_count = 0;
    for (int i = 0; i < 1000; ++i)
    {
        if (a[i] > 0)
        {
            positive_sum += a[i];
            positive_count++;
        }
    }

    if (positive_count > 0)
    {
        double positive_average = positive_sum / positive_count;
        std::cout << positive_average << '\n';
```

```
    }
    else
    {
        std::cout << "No positive values in the array.\n";
    }

    return 0;
}
```

Q2. You are given the following array (which is sorted in ascending order):

-8	-7	-7	-3	-1	1	2	3	4	4	8	12	15	16	19

Using the binary search algorithm from our notes, list down the index values and the corresponding array values of the array that was actually accessed during the search for the target value of 2. (I have already completed the first row for you.)
Answer:

```
7 3 5 6
```

Q3. Refer to the above question. How many probes (i.e., number of times a value in the array was accessed) did the above binary search make?
Answer:

```
4
```

Q4. Refer to the array in the above question. How many probes do you need if you use linear search to find 2?
Answer:

```
7
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.